Prismatic Analysis of AMR Interpretive Divergence

Step 0: Pre-Analysis Reflexive Declaration

- **Likely home grammar:** My intuition and background align with the **Global Public Health/One Health** perspective. I naturally think in terms of health outcomes, disease burden, and multisector coordination.
- **Hardest grammar to operate:** The **Pharmaceutical/Economic** grammar will be hardest. Its emphasis on market incentives, investment flows, and profit motives is far from my usual focus. I worry I might underestimate how industry dynamics shape AMR signals.
- **Possible blind spots:** Given my public-health orientation, I might underappreciate agricultural concerns (livelihoods, food security) and understate the importance of economic feasibility. I could miss how profitability and industry structure drive real-world outcomes.
- **Temporal orientation:** I am inclined to **causal-historical** analysis (looking at how past practices created current AMR) and also to **prescriptive/policy** thinking. I focus on chains of cause and effect, but also on recommending interventions for the future.

Section 1: Observation Field Declaration

- **Phenomenon, frame, scope:** The case study is the *global response to antimicrobial resistance (AMR)* during 2015–2026. This involves international policies, national programs, and sectoral actions. It spans **one-health** realms (human, animal, environmental health) and is inherently global in scale (WHO, OECD, FAO, etc.).
- **Why interpretive divergence:** Although AMR is presented as a "consensual crisis," different communities define the problem and solutions differently. For example, some focus on human mortality and stewardship (health community), others on innovation incentives (economic/industry), and others on farming practices and food security (agriculture). These lenses pull the same signals into divergent narratives. The contested nature of many signals (cause attribution, cost estimates, animal vs human drivers) means people **disagree on what the data "means"**. This hidden divergence emerges from different causality assumptions and priorities.
- **Signal contestation level:** This domain is a **mixed** signal domain. Many signals (death estimates, resistance rates, usage trends) are widely cited, but how they are interpreted is contested. Some signals (e.g. economic models, agricultural usage) are explicitly disputed or downplayed by certain actors. We will mark particularly **contested signals** (below).
- **Major interpretive communities:** We identify at least three broad communities (grammars):
    - **Global Health/One Health community:** WHO, UN agencies, health ministries, public-health researchers and NGOs. They frame AMR in terms of disease burden and multisector coordination.
    - **Pharmaceutical/Economic community:** Pharma companies, investors, OECD analysts, health economists. They view AMR through market and innovation lenses (pipeline, incentives, cost models).
    - **Agricultural/Food Production community:** FAO, OIE (WOAH), farming industry, agricultural ministries. They emphasize livestock antibiotic use, food security, and farming practices.

Section 2: Signal Set

We list **core signals** (type, source, ambiguity):

- **Global AMR mortality (2019):** ~1.27 million deaths directly attributable to resistance; ~4.95 million associated

; projected +52.3% (to ~75.1 billion) by 2030 if unchecked ; projected +8% to ~107,472 tonnes by 2030. Major consumers: China (~32,776 t), Brazil, India, USA, Australia (combined ~58% of usage) , up from 80 in 2021; but only ~12 are truly "innovative" against priority pathogens. Since 2017, 13 new antibiotics were approved but only 2 represented new drug classes), and several major pharma companies (Novartis, Sanofi, AstraZeneca) abandoned antibiotic programs. Two biotech companies with new antibiotics went bankrupt (Melinta and Achaogen) . In 2022, 60 countries reported national antibiotic use data to GLASS. However, WHO notes "serious data gaps" in many low-income regions . Many have established stewardship programs and use WHO's AWaRe antibiotic classification. GLASS data indicate many countries still fall below the UN target of ≥70% 'Access' antibiotics usage (≈$29B health system cost + $37B productivity loss). The World Bank (2016) projects in a high-impact scenario that by 2050 AMR could shrink global GDP by ~3.8% and cumulatively cost ~$100 trillion

- . Nonetheless, actual "One Health" implementation varies by country and is often more rhetorical than operational. *(Qualitative; policy documents. Ambiguity: "One Health" means different things; adoption level is debated.)*

Counter-signals: We note several signals used to question or qualify the above: (1) *Mortality estimate critique:* Some commentators argue the Lancet estimates may over-count AMR deaths due to confounding; alternative models yield lower figures. (2) *Animal-use impact debate:* Critics suggest that human AMR is driven mostly by medical use, not agriculture, pointing to uncertain zoonotic transmission; they downplay animal-use signals. (3) *Economic model skepticism:* Economists warn that projected GDP losses ($66 B/yr or $100 T total) assume worst-case scenarios and may not account for adaptation; some see these as alarmist "ballpark" figures. (4) *Pipeline optimism:* Industry notes the pipeline is growing (97 agents) and highlights new business models (non-traditional therapies) as emerging, questioning narratives of total shortage. (5) *One Health skepticism:* Some view One Health as well-intentioned rhetoric lacking clear metrics, so signals of "One Health plans" can be discounted as gestures.

Contested Signals (examples):

- *AMR mortality burden:* Public Health grammar treats 1.27 M deaths (2019) as a baseline fact justifying urgent action. Economic grammar may cite the larger 4.95 M "associated" figure but stress uncertainty and focus on economic losses instead. Agricultural grammar regards the human death count as largely outside its remit (most deaths are from human pathogens in hospitals, not farm-related).
- *Veterinary antibiotic use:* Public Health grammar sees 99,500 t (2020) as evidence that agriculture drives resistance. Economic grammar treats it as an input to total consumption but focuses on cost impacts. Agricultural grammar emphasizes that this usage reflects essential practices for food production; it contests whether reducing it is feasible without harming output.

Section 3: Grammar Specifications

We define three grammars with their parameters:

Grammar A – Global Public Health (One Health) Grammar:

- **Primary analytical variable:** *Disease burden and infection prevalence* (e.g. AMR morbidity/mortality).

- **Signal selection rules:** Admits signals related to health outcomes, resistance rates, infection incidence, consumption in humans and animals (as one health). Prioritizes WHO data, Lancet studies, epidemiological projections, surveillance coverage.
- **Weighting priority:** Highest weight to human mortality/morbidity estimates and hospital infection data; next to resistance prevalence; then to consumption trends and stewardship coverage. Economic signals are considered insofar as they affect health budgets.
- **Causal chain type:** Multi-causal, involving ecological and social links (misuse in humans/animals → rising AMR → more infections and deaths). Primarily *backward- and forward-causal*: it traces causes (past prescribing, sanitation) and also uses projections to predict outcomes of current trends.
- **Admissible closure:** Conclusions must focus on health interventions: strengthening surveillance, infection control, stewardship, R&D, and multisector coordination. Closure is restricted to interventions that can be justified by population health outcomes (e.g. "we must invest in X to save lives"). **Admissible closure is broad** but framed in public-health terms; it does not include rejecting the crisis or purely economic solutions.
- **Excluded signals (Rule 6):** Signals that center on market profitability or sector-specific economics (like quarterly pharma profits, stock prices) are de-emphasized. Direct farm-output metrics (e.g. yield loss) are secondary. Signals about individual patient cost (not systemic) or equity are secondary.
- **Characteristic vocabulary (Rule 7):** Terms like *"surveillance," "stewardship," "infection control," "infection incidence," "antibiotic-resistant infections," "community spread," "universal health coverage," "One Health," "mortality," "equity."* Rhetoric emphasizes *"prevent," "strengthen capacity," "global action," "urgent," "contain," "safe use."*
- **Temporal orientation (Rule 8):** Primarily *causal-historical* (analyzes how past use caused present AMR) and moderately *prescriptive* (advocates future policy to prevent death). It uses projections for planning but focuses on the historical trajectory of resistance as lesson.

Grammar B – Pharmaceutical/Economic Grammar:

- **Primary analytical variable:** *Market incentives and R&D dynamics*. The grammar asks: "What economic forces govern antibiotic development?"
- **Signal selection rules:** Admits signals on pharmaceutical pipelines, company investments, market exits, patent policies, health economic models. Prioritizes industry data, OECD cost-benefit studies, grant funding levels, policy incentives (e.g. push/pull programs). Clinical efficacy and epidemiological signals are noted only insofar as they affect market viability.
- **Weighting priority:** Highest weight to the number of drugs in clinical/preclinical pipeline, R&D expenditure, regulatory policies (gain act, market entry rewards), and economic ROI calculations. Secondary weight to healthcare cost projections (as justification for spending). Low weight to raw mortality counts or farm usage.
- **Causal chain type:** *Econometric-causal*: "Market failure (lack of profit) → insufficient innovation → sustained AMR problem → economic losses." It is largely forward-looking/prescriptive, focused on cause-effect in markets (e.g. "absence of incentives *causes* R&D collapse; introducing incentives *causes* pipeline revival").
- **Admissible closure:** Closures focus on economic and regulatory solutions: implementing incentives, altering patent rules, funding R&D consortia, and building profitable business models. Conclusions about restricting use or public health education are only admissible if they tie back to market efficiency (e.g. stewardship as preserving drug value). Political or equity-based conclusions are beyond this grammar's scope unless cast in economic terms.
- **Excluded signals:** Direct human life-toll (except as cost), environmental signals, and ethical arguments are largely excluded. Signals about animal/veterinary usage are treated as side effects. Basic scientific discovery is not enough; must tie to commercialization.
- **Characteristic vocabulary:** *"Pipeline," "push/pull incentives," "market entry reward," "R&D funding," "return on investment (ROI)," "cost-effectiveness," "patent exclusivity," "subscription model," "push*

grants," "pull rewards," "reimbursement policy." Rhetoric tends to be pragmatic: *"sustainable business models," "market failures," "cost-benefit," "productivity," "innovation ecosystem."*

- **Temporal orientation:** Strongly *prescriptive/forward-oriented*. It uses projections to justify future policy steps and focuses on long-term trends in innovation funding. Causal analysis is used to design interventions (forward use) rather than purely reconstruct past usage.

Grammar C – Agricultural/Environmental (One Health) Grammar:

- **Primary analytical variable:** *Livestock production and ecosystem health.* It asks: "How does antibiotic use impact animal health, agriculture, and environments?"
- **Signal selection rules:** Admits signals on veterinary antibiotic consumption by sector and country, animal infection rates, farm yields and productivity measures, trade data (e.g. restrictions on exports), and animal-health R&D (e.g. vaccines). Also considers environmental AMR data (e.g. contamination of water by farm runoff). Human clinical signals are admitted only if directly linked to agriculture.
- **Weighting priority:** Highest weight to agricultural indicators (e.g. tonnes of antibiotics in feed, counts of animal disease outbreaks, cost of livestock loss). Next to ecosystem impacts. Lower weight to human morbidity/mortality numbers. Economic indicators are considered insofar as they affect farmers (food prices, income).
- **Causal chain type:** *Agro-ecological and systems-oriented.* It often reasons *laterally* (balancing competing needs) or *forward-causal* within the agro-ecosystem. The chain: "Antibiotic use in animals → resistance in animal reservoirs and environment → spillover to human or lower animal productivity." But also: "Restricting use → changes in yield and farming practices." It tends to highlight feedback loops in ecosystems and markets.
- **Admissible closure:** Solutions are focused on agriculture: tighter veterinary regulation, farm biosecurity, alternative practices (vaccines, probiotics), and environmental safeguards (waste treatment). It may endorse One Health coordination, but only if framed as supporting farming (e.g. integrated surveillance to protect livestock). Purely human-health measures (like hospital stewardship) are excluded unless they have indirect agricultural benefit.
- **Excluded signals:** Pure human-clinical metrics (like hospital MRSA rates) are downplayed. Signals that prioritize public health over food supply (e.g. mandatory antibiotic bans without support) are disfavored. Signals on pharmaceutical profitability or hospital costs are irrelevant unless they tie back to agriculture.
- **Characteristic vocabulary:** *"Livestock," "herd health," "biosecurity," "poultry/swine/pet systems," "growth promotant," "vaccine coverage (in animals)," "feed additives," "food security," "export markets," "antibiotic withdrawal periods."* Emphasis is on *"sustainable agriculture," "animal welfare," "farmer incomes," "production targets."* The tone often stresses *"resilience," "balance," "multifunctional landscapes," "ecosystem health."*
- **Temporal orientation:** Largely *lateral/intersectoral* with some *forward-leaning* planning. It considers how human and animal systems co-evolve (interpreting signals in their broader agronomic meaning). Solutions are often generational (improve practices over time) rather than immediate crisis responses.

Grammar boundaries:

- *A → B:* If Grammar A (Health) shifted its primary variable from **human health outcomes** to **economic returns**, and started treating drug development costs as fundamental, it would become Grammar B. In practice, down-weighting mortality in favor of ROI and profit concerns would convert the public-health grammar into a market-based grammar.
- *A → C:* If Grammar A instead focused on livestock metrics (herd infection rates, yields) and began excluding human clinical data, it would become Grammar C. E.g., interpreting the same evidence but asking "What does this mean for animals and agriculture?" instead of "What does this mean for people?" marks that boundary.

- *B → C:* If Grammar B (Economic) stopped focusing on market incentives and redefined the core problem as one of agriculture, re-weighting signals like veterinary use and food prices above pipeline counts, it would become Grammar C. That boundary is crossed when incentives-based reasoning gives way to farming-centric reasoning.

Source traditions:

- Grammar A draws from WHO/UN publications, Lancet and JAMA Infectious Diseases articles, CDC/NIH epidemiology, global health NGO reports, etc.
- Grammar B uses OECD reports, World Bank analyses, WHO economic briefs, pharmaceutical trade association statements, pharmaceutical R&D dashboards (e.g. Pew/WHO pipeline reports), and economic journals.
- Grammar C references FAO/OIE reports, veterinary scientific literature, national agriculture department data, environmental AMR studies, and commodity trade analyses.

Section 4: Signal Processing

Grammar A: Global Public Health Grammar

From the **Global Public Health** perspective, AMR is defined by its human health impact. Core signals are interpreted around disease burden: for example, the Lancet's estimate of ~1.27 million annual deaths directly attributable to AMR in 2019 (4.95 M including associated cases)

is treated as a catastrophic warning. Rising antibiotic use in people (+16.3% from 2016–2023) is read as fueling this crisis; veterinary antibiotic use (~99,500 t in 2020) is also weighted heavily, reinforcing that animal use contributes to human infections under a One Health view. Data gaps (e.g. "serious data gaps in many low-income countries") are acknowledged but interpreted as meaning the true burden might be even higher than reported. Economic cost models (OECD's ~$66 B/year

; World Bank's $100 T by 2050) are cited to underscore urgency, particularly for vulnerable nations. In Grammar A, signals like mortality and resistance rates are **top-weighted**, while sales figures or profits are virtually ignored.

The causal chain under this grammar runs: *antimicrobial overuse (in humans and animals) → rising resistance rates → more untreatable infections → higher mortality/morbidity and strained health systems*. It mixes historical analysis (how past prescribing and sanitation led here) with forward projection (e.g. resistance to last-line drugs may triple by 2035

). Projections (like +52% antibiotic consumption by 2030) are read literally as potential futures to avoid. Interventions (stewardship, sanitation, vaccines) are justified both by past success (e.g. HIV and TB drug development as precedents) and by economic ROI (investing ~$4 per person/year in One Health could avert thousands of deaths

).

Counter-signals are integrated contextually. If skeptics claim the 1.27 M death figure is inflated, Grammar A replies that even *half* that number is huge (on par with malaria/HIV)

. If industry points to *97* pipeline drugs in development, this grammar notes that only ~12 are truly novel

, so "the pipeline is nowhere near enough" to offset resistance. Claims that cutting farm antibiotics would collapse food supply are met by pointing to alternatives: e.g., phased reductions, improved hygiene, and veterinary vaccines can mitigate the impact while controlling resistance. In each case, the grammar

re-frames challenges in public-health terms: human life is the overriding metric, so even strong economic arguments are weighed against the health cost of inaction.

The closure of Grammar A is a comprehensive health-sector program. It endorses global coordination (all 178 countries now have AMR action plans

) and expanded surveillance (GLASS participation in 127 countries). It calls for strict stewardship (e.g. meeting WHO's target that ≥70% of antibiotic use be "Access" drugs) and infection prevention (clean water, vaccinations). The grammar also emphasizes sustained R&D funding – invoking WHO's objective to "increase investment in new medicines, diagnostic tools, vaccines"

– and public-private partnerships (e.g. GARDP, AMR Action Fund). In sum, Grammar A concludes that **multisector One Health action** is essential: a package of prevention, rational use, and innovation aligned with WHO/OECD recommendations. This closure is the product of the grammar's architecture – by prioritizing population health signals and One Health integration, it yields a public health–centric policy agenda (not my personal preference, but the inevitable output of this interpretive frame).

Grammar B: Pharmaceutical/Economic Grammar

From the **Pharmaceutical/Economic** perspective, AMR is framed as a market/investment issue. Core signals are financial and pipeline metrics. For instance, this grammar sees the **scarcity** of profitable antibiotics as key: 97 candidates in clinical development (2023)

is treated as far too low, especially since only ~12 are innovative. It emphasizes the decline in industry commitment: Alliance members' antibiotic R&D investment fell from $2.0 billion in 2017 to $1.6 billion in 2018, and major firms (Novartis, Sanofi, AstraZeneca) have exited antibiotics entirely. Even companies that achieved approval have gone bankrupt (Melinta, Achaogen)

. These signals – pipeline count, R&D spending, company exits – are given highest weight. Human death counts are not ignored, but mainly to quantify market potential (e.g. "X patients need drugs"). Animal antibiotic totals are largely treated as another market segment (potential demand) rather than a driver of the human problem.

The causal chain in Grammar B is econometric: *insufficient profitability → insufficient R&D → inadequate supply of new drugs → AMR remains unchecked → economic losses*. It is primarily forward-looking, focusing on policy fixes: e.g. if market-entry rewards are implemented, then R&D will increase. Economic models are core evidence: OECD finds AMR costs ~$66 B annually

; the World Bank warns of multi-trillion-dollar losses. Grammar B sees these not as abstract tragedies but as justification for investment. It calculates return on investment (e.g. every $1 spent averts $5 in losses). For this grammar, every signal is translated into dollars. For example, a mortality statistic becomes "X dollars in healthcare and productivity loss," which then informs how much incentive is justified. Predictions of rising resistance (e.g. doubling by 2035

) are interpreted as future market needs: we must create an economic model that ensures new antibiotics are developed before those deaths occur.

Counter-signals are recast as market feedback. If public-health voices demand drastic cuts in all antibiotic use, Grammar B counters that overly restrictive policies would shrink the market, deterring investment. If it hears that the AMR death toll is enormous, it agrees the social cost is high but emphasizes that without a revenue model, no new drugs will come. If reports claim the pipeline is stagnating, Grammar B may highlight recent growth (97 up from 80) or point to emerging therapies (phage products, biologics) as expanding the market beyond traditional antibiotics. The principle is: signals are used to adjust economic levers. For example, low global sales of new antibiotics (due to conservation) is not a failure of the drugs,

but a sign that *revenue models must change*. Resistance prevalence is seen as a signal that prescribers are withholding antibiotics (good practice) – and that pharma must be paid differently (e.g. subscription models) to remain viable.

The closure of Grammar B is a set of market-based reforms. The conclusion is that the **market must be fixed** to solve AMR. Specific prescriptions include "push" incentives (grants, subsidies for R&D) and especially "pull" incentives (market-entry rewards, transferable exclusivity, subscription payments) to guarantee returns. The grammar might endorse approaches like the U.S. GAIN Act or the proposed subscription models in Europe, arguing they align profit with public good. It may support partial stewardship (to extend drug lifetimes) but primarily as a safeguard of value. In essence, Grammar B's closure is: *make new antibiotics pay*. Unlike Grammar A's broad health program or Grammar C's farming measures, this outcome is narrowly focused on financing and regulation. Again, this result stems directly from the grammar's logic: by treating AMR as a corporate/investment issue, its solution space is constrained to economic incentives and industry policy (not a moral judgment, but the architecture's inevitable output).

Grammar C: Agricultural/Environmental Grammar

From the **Agricultural/Environmental** perspective, AMR is framed as a challenge for livestock and ecosystems. Core signals here are agricultural: for instance, the ~99,500 tonnes of antimicrobials used in food animals in 2020

. This grammar highlights that livestock accounts for a majority of global antibiotic tonnage (China, Brazil, India, USA, Australia alone contribute ~58% of use

). These consumption figures imply to Grammar C that "antibiotics are essential to modern farming." Therefore, signals about crop yields, animal disease incidence, and regulatory changes (e.g. bans on growth promoters) would also be focal if available. Human mortality signals (e.g. 4.95 M associated deaths) are noted but rank lower – this grammar interprets them through the lens of farming (asking: *are these human cases linked to agriculture?*). Essentially, the primary concern is how AMR policies affect animal health and food security.

The causal view in Grammar C emphasizes farm-level systems: *routine antibiotic use in herds → selection of resistance in animals and the environment → possible transmission to humans.* But it also factors the reverse causal importance: restricting antibiotics too fast could harm animal welfare or yields. Therefore it evaluates interventions by both health and agricultural criteria. For example, it might cite the EU's experience: after banning antibiotic growth promoters, animal use and resistance dropped without catastrophic loss in output (though adjustments were needed). This grammar uses data comparatively: e.g. "in country X, reducing farm use by Y% was achieved with Z interventions." Lacking robust AMR data, it often reasons by analogy (veterinary success stories) and economic cost to farmers. The approach is somewhat lateral/hermeneutic: interpreting AMR as a complex socio-ecological problem where human and animal interests must be balanced.

Counter-signals are absorbed skeptically. If told "AMR will kill millions of people," Grammar C may respond that most current deaths are due to hygiene and healthcare gaps, not farm antibiotic use. It might downplay One Health mandates as "we must include farming concerns." Conversely, signals about food security hit home: e.g. a World Bank report projects a 2.6–7.5% annual decline in global livestock output by 2050 under high AMR scenarios

. Grammar C treats this as a direct threat to food supply. It also notes any evidence of effective alternatives: if an animal vaccine eliminates a common livestock disease, that is highly welcome. In general, this grammar treats signals as indicating how to achieve "healthy animals and healthy profits." If economic models focus on healthcare loss, Gram C reframes them into agricultural terms (lost farm productivity, income).

The closure of Grammar C is an agronomic action plan. It supports prudent, targeted measures: phasing out non-therapeutic (growth-promotion) antibiotics, enforcing veterinary prescriptions, and improving farm hygiene/biosecurity. It calls for investment in animal vaccines, better breeding, and alternative disease controls (e.g. probiotics, prebiotics). It also advocates for consumer incentives (labeling antibiotic-free products) and compensation for farmers adapting to new rules. The environment is safeguarded by waste-treatment signals. Essentially, Grammar C's solution is **sustainable agriculture**: reduce AMR risk while keeping food production stable. This outcome follows directly from its focus on farm signals – it would not emerge from purely health- or finance-based reasoning. In other words, the Agricultural grammar's closure is a product of its architecture: its metrics and vocabulary inevitably lead to recommendations centered on livestock and eco-systems rather than global population health or market reforms.

Section 5: Closure Comparison

- **Closure content:** Grammar A yielded a *broad public-health program* (global surveillance, stewardship, R&D funding, One Health coordination). Grammar B produced a *market/incentive package* (economic policies, funding mechanisms for drug development). Grammar C produced an *agricultural resilience plan* (regulated animal use, farm biosecurity, animal vaccines). Thus, A's closure emphasizes health interventions, B's emphasizes financial incentives, and C's emphasizes farming practices.
- **Signal divergence:** The three grammars each prioritized different signals. Grammar A elevated human mortality and infection rates

; B elevated R&D pipeline and investment metrics; C elevated veterinary usage figures and hospital infection data were important to A but irrelevant to C. Conversely, C's focus on livestock outputs (absent from WHO sources) was outside A's and B's concerns. Each grammar "leaked" little of the others' emphasis: B acknowledged some health cost figures (since they inform ROI) and A at times cited OECD cost data

- , showing minor cross-over, but on the whole each grammar filtered out the others' domains.
- **Causal architecture:** All grammars claimed a causal chain but of different form. Grammar A's chain was *multi-factorial epidemiological* (overuse → resistance → mortality). Grammar B's was *econometric* (market failure → no new drugs → economic loss). Grammar C's was *agro-ecological* (animal use → resistance → potential output loss). Notably, A and B share a one-way cause (abuse → harm) but differ in "harm" (human deaths vs economic cost). Grammar C's causality is bidirectional: it cares about both cause (farm use causes some AMR) and effect on farming viability. The shape of explanation differs: A used historical trends to justify future action, B used cost–benefit projections to design incentives, C used comparative analogies (e.g. EU vs non-EU farming outcomes) to guide transitions.
- **Closure range:** Grammar A produced the **widest closure**, calling for action across health, agriculture, finance. Its solution space was broad: it effectively encompassed elements of B and C (since one-health). Grammar B's closure was **narrower** – focused on creating an innovation ecosystem and market reforms; many public-health details were not actionable within its view. Grammar C's closure was **sector-specific**, centered on agriculture and environment. Timing and confidence also differ: A's closure was urgent and expansive ("we must do many things now"), B's was longer-term and technical ("we need economic reforms which take time"), C's was gradual and balanced ("we can't stop feeding people overnight").
- **Closure timing:** Grammar A reached its conclusions **quickly and confidently**, since it treats AMR as an immediate crisis; it gave concrete steps right away. Grammar B was somewhat more tentative, emphasizing that changes in incentives take legislative and corporate action over years. Grammar C was cautious and condition-based: it stressed adaptation steps, acknowledging that all measures must be calibrated to avoid unintended harms. In practice, A seemed the most unequivocal, C the most pragmatic about pacing, and B the most deferential to future planning.

- **Ontological check:** Do they analyze the "same" phenomenon? Partially. Grammars A and B mostly define AMR as a crisis in human health (A by lives lost, B by economic impact), whereas Grammar C effectively sees AMR as partly a *different problem* – an animal health/agro-environment issue. They share the label "AMR," but C's primary lens is livestock and environment, not human patients. There is some ontological shift: Grammar C conceives the problem's core object as *"antibiotic reliance in food systems,"* whereas A/B see *"resistant infections in humans"*. Thus the grammars overlap but are not fully coextensive in object; each constitutes AMR in terms that others might consider only part of the picture.

Section 6: Counter-Signals, Leakage, and Stress Test

- **Consolidated counter-signal analysis:** Throughout the analysis, we integrated counter-signals into each grammar's reasoning (see processing sections). Consolidating: Public Health grammar had to absorb signals that emphasized economics or industry (which it downplayed); Pharma grammar absorbed health-driven criticisms; Agricultural grammar absorbed human-centered data as externalities. In summary, each grammar flagged its own threats: e.g. Grammar A noted that industry warnings about the pipeline were too optimistic; Grammar B noted that stewardship concerns might conflict with market goals; Grammar C highlighted uncertainty in attributing human cases to animal use. Where signals were "contested," each grammar interpreted them defensively in its own frame (see Section 4 for examples). No grammar uniformly accepted all signals at face value; each treated the others' alarm bells as secondary.

- **Leakage instances:** Some signals naturally "leaked" between grammars: e.g. Grammar A (public health) invoked the OECD ROI analysis

– a typically economic argument – to strengthen its case for One Health investment. Grammar B (economic) reluctantly recognized public-health data, such as the human mortality cost estimates, in framing AMR as an economic threat. Grammar C (agricultural) had to acknowledge World Bank economic impacts on livestock

- . These leakages show that no grammar can be perfectly insulated: each borrowed a few elements from others when they were rhetorically convenient. However, the core assumptions remained distinct.

- **Grammar Stress Test:** For each grammar, we identify one hypothetical "breaking" counter-signal:

    - *Grammar A (Public Health):* **Threatening counter-signal:** Suppose new data convincingly showed that human AMR infections are plummeting due to an unrelated factor (e.g. a global rollout of an effective vaccine or revolution in sanitation), making deaths negligible. This undermines the "crisis" premise. Can Grammar A absorb it? In that case, its emphasis on AMR as a human health emergency would collapse; its closure (urgent broad action) would be untenable unless it shifts focus entirely (perhaps to TB/HIV as "also AMR" and starts again). Without the mortality problem, Grammar A would need to re-frame AMR as a chronic manageable issue.

        - *Assessment:* This is catastrophic for Grammar A's closure. Its confidence depends on high burden; if that burden vanished, the architecture yields a very different, likely much weaker conclusion. Grammar A's closure is thus conditional on the high-burden premise.
    - *Grammar B (Economic):* **Threatening counter-signal:** Imagine a revolutionary non-profit antimicrobial discovery platform suddenly made multiple new, cheap, broad-spectrum antibiotics (or bacteriophage cocktails) available, effectively outcompeting the traditional

market model. If drugs could be developed without profit motives, the market failure is sidestepped. Grammar B must ask: does a solution exist outside market mechanisms? It might admit success but maintain that such a scenario *itself* confirms the need for alternative incentives (since traditional R&D wasn't able to do that). However, the underpinning assumption (that only profit can drive innovation) would be shaken.

    - *Assessment:* Grammar B can partially absorb this by reframing it as showing that when incentives align, innovation happens; but it would struggle to conclude the same policy recommendations (it wouldn't need extra incentives if a non-market model worked). Thus Grammar B's closure (based on creating incentives) would have to adapt to the new context. Grammar B's conclusion is somewhat conditional on traditional market constraints remaining.
  - *Grammar C (Agricultural):* **Threatening counter-signal:** Suppose an independent global study proved almost no transmission of human-resistant infections from farm animals (i.e. an "AMR disconnect"). If animal antibiotic use had been found not to affect human AMR, Grammar C's premise (that agriculture is a major source of AMR) is undermined. However, Grammar C's closure (prudent use plus alternatives) primarily focused on farm resilience, so it might simply downplay the need for change. It could then say: "If animals aren't causing human AMR, then continue essential use with basic oversight." Its arguments for replacing growth promoters or enhancing farming might lose justification.

    - *Assessment:* This counter-signal would erode Grammar C's impetus for strict agricultural measures. The grammar might still endorse moderation, but its rationale weakens. Unlike Grammar A, Grammar C does not *depend* on cross-species AMR for closure, so it can absorb it by shifting the justification (e.g. focus on resistance in animals themselves, or trade restrictions that arise anyway). Grammar C's conclusions are somewhat robust to this counter-signal, though some justification is lost.

Each grammar's closure is thus potentially overturned or necessitates re-formulation by its most threatening counter-signal. Grammar A is most fragile to a drop in human AMR; Grammar B to non-market innovation; Grammar C to evidence negating agricultural impact. When a grammar cannot absorb its stress-test signal, we see that its original closure was **conditional** on certain assumptions (health burden, market failure, agro-impact).

Section 7: Post-Analysis Reflexive Note

Reflecting on my Step 0 predictions, my home grammar (Public Health) did indeed dominate my initial perspective, and Grammar A's analytic lens came most naturally. The hardest to operate was indeed the Pharmaceutical/Economic grammar: I frequently had to consciously shift from health-emphasis to an investor mindset, which was challenging. I also found the Agricultural grammar somewhat foreign, as I had less intuition about farming systems. A blind spot I hadn't fully anticipated was underestimating the extent to which economic models abstract human lives into GDP figures; I had to be careful to speak in that jargon when simulating Grammar B. If I had operated from a different home grammar (say, the economic lens), I suspect I would have constructed very different questions from the start, perhaps ignoring raw mortality signals altogether and focusing on investment flows. This exercise shows that interpretive architectures deeply shape the questions analysts ask and the solutions they see as viable: the same evidence yields very different "truths" depending on the framing. It highlights how our own background biases signal selection and weighting; being aware of my bias helped me to deliberately incorporate missing perspectives. Ultimately, the exercise reveals that AMR is not a single unified "object" but a contested arena where evidence is channeled by ideological and disciplinary grammars into divergent conclusions.

Section 8: Methodological Observations

Applying the prismatic method (v2.0) worked well to systematically reveal hidden divergences. Integrating counter-signals throughout (rather than tacking them on) forced me to see how each grammar naturally neutralizes opposing views. The temporal orientation parameter was useful: noting whether each grammar was backward- or forward-looking clarified why some emphasized historic data versus projections. Defining clear grammar boundaries helped articulate the core difference between, say, health vs market frames. The **stress test** was particularly revealing: it showed which core assumptions each grammar could not accommodate, highlighting their conditionality. I encountered difficulties balancing comprehensiveness with focus: each grammar lens has a huge literature, so picking representative signals was hard. Also, ensuring not to conflate *my* views with the grammar's was tricky (I had to keep phrasing in terms of the grammar's logic). The new features (counter-signals, temporal orientation, boundaries, stress tests, ontological check) were all helpful; they forced deeper reflection on each grammar's architecture. For instance, the ontological check made me realize that Grammar C almost treats "AMR" as a different phenomenon (animal vs human). For future refinement, clearer guidance on writing grammar-specific essays (e.g. first-person "the grammar would say X") might help avoid ambiguity. Overall, v2.0's structure was effective in eliciting a thorough, multi-angled analysis, though executing all parts in detail is quite demanding.

. *(Quantitative; sourced from the Lancet study via WHO; model-based estimation. Ambiguity: relies on assumptions and incomplete data in LMICs, so some dispute the exact numbers.)*
Human antibiotic consumption trend: Global defined-daily-doses (DDDs) increased ~16.3% from 2016 to 2023 (29.5 → 34.3 billion DDD)
. *(Quantitative; WHO/OECD estimates. Ambiguity: projections assume current trajectories; COVID-19 effects complicate trends.)*
Animal (veterinary) antibiotic use: ~99,502 tonnes of active ingredient used globally in food animals in 2020
. *(Quantitative; data from global modelling with wide confidence intervals. Ambiguity: OIE data are coarse, and some countries lack reporting; actual patterns are debated.)*
Antibiotic R&D pipeline: WHO reports ~97 antibacterials in the global pipeline (2023)
. *(Quantitative; WHO surveys. Ambiguity: criteria for "innovative" vary; industry notes many candidates are derivatives.)*
Industry investment and exits: The AMR-focused biotech alliance reports R&D investment fell sharply (Alliance members' antibiotic R&D spend was $2.0B in 2017, $1.6B in 2018
. *(Qualitative/Quantitative; news reports and industry surveys. Ambiguity: some small companies persist, but overall trend is considered dramatic.)*
Surveillance coverage: WHO's GLASS surveillance system had 127 countries enrolled by end of 2022
. *(Quantitative; WHO reports. Ambiguity: enrollment does not imply data quality or representativeness; some regions under-report.)*
Stewardship and National Action Plans: As of late 2023, 178 countries have adopted AMR National Action Plans aligned with WHO's Global Action Plan
. *(Mixed; indicator-based. Ambiguity: "having a plan" does not equal implementation; AWaRe targets may not reflect clinical realities.)*
Economic cost estimates: OECD estimates AMR costs ~USD 66 billion (purchasing-power-parity) annually across OECD/EU countries
. *(Quantitative; model-based. Ambiguity: depends on scenario assumptions about healthcare adaptation and global trade.)*
One Health framework adoption: International bodies have embraced One Health. In 2022 the Quadripartite agencies launched a One Health Joint Plan of Action addressing AMR among other threats
. Signals like NAP adoption or vaccine development were strong in A, moderate in B (only as economic projects), and peripheral in C. Economic cost models strongly influenced A and B but not C (which saw them as distant). Conversely, C took veterinary surveillance and productivity signals as critical (though we had few explicit sources on these); A largely ignored those, and B treated them only as market size. This

divergence shows how each grammar's *signal selection and weighting* reshape the narrative: the same data can vanish or dominate depending on the lens.

Exclusion patterns: Grammar A effectively excluded corporate profit signals and most farm/economic data. Grammar B excluded most public-health details (like death rates and stewardship successes) unless monetizable. Grammar C excluded health-care settings entirely and largely ignored formal economic incentives. For example, AWaRe targets